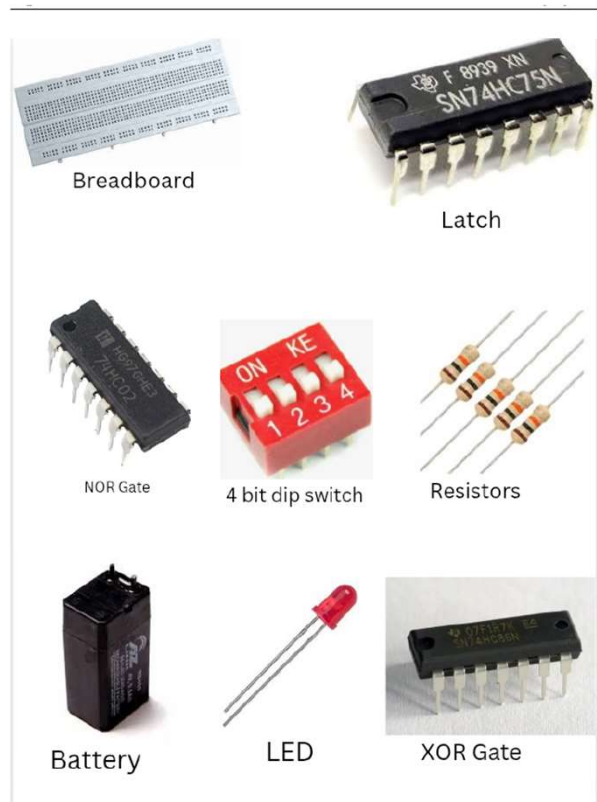


## 4-bit Password Security system (using Latch , XOR & NOR);

This idea is of this system is that 2 Dip switches used one Dip Switch **“Key Code Switch”** to enter/change fixed password (correct password) and other Dip Switch **“Data Entry Switch”** use to enter the password to unlock the lock. The “Key code switch” enter the password to **Latch** where password get stored when enable of Latch is at **HIGH** and when enable is LOW then Latch uses stored password even if “key code switch” value gets change. Here we use XOR Gate which compares both password and gives output. If someone is trying to open the lock enters the code at “Data Entry Switch” which is similar to the stored password of Latch then **Green LED** will light up and the lock will open while, if he enters the wrong password then the **RED LED** will light up which is an indication that some wrong person is unlocking the lock.

# Materials Required :

- 7475 Latch
- 7486 XOR Gate
- 7402 NOR Gate
- Two four – position DIP Switches
- Two Light Emitting Diode(Red ,Green)
- Four 1N914 Diode
- Four 1K $\Omega$  Resistor
- Four 10K $\Omega$  Resistor
- Keylock switch
- 4V Battery
- Breadboard
- Male-male Jumper wires



Jumper wires



Keylock switch



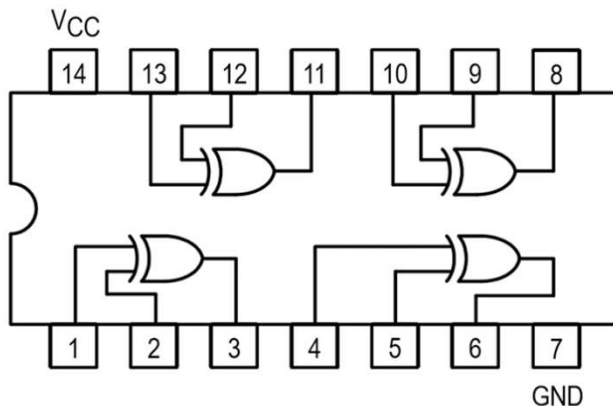
# Truth Table & Pin Diagram

## 1. XOR Gate (Exclusive OR Gate)

**Symbol:**  $\oplus$

**Function:** Output is **HIGH (1)** if the inputs are **different**, else it is **LOW (0)**.

| INPUT |   | OUTPUT |
|-------|---|--------|
| A     | B | C      |
| 0     | 0 | 0      |
| 0     | 1 | 1      |
| 1     | 0 | 1      |
| 1     | 1 | 0      |

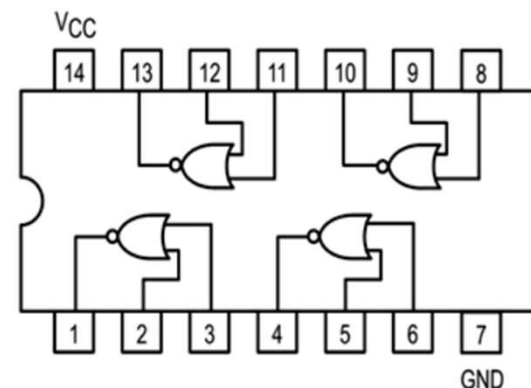


## 2. NOR Gate (NOT-OR Gate)

**Symbol:** OR gate with a small circle (inversion bubble) at the output.

**Function:** Output is **HIGH (1)** only when **all inputs are LOW (0)**.

| Input | Input | Output |
|-------|-------|--------|
| A     | B     | Y      |
| 0     | 0     | 1      |
| 0     | 1     | 0      |
| 1     | 0     | 0      |
| 1     | 1     | 0      |



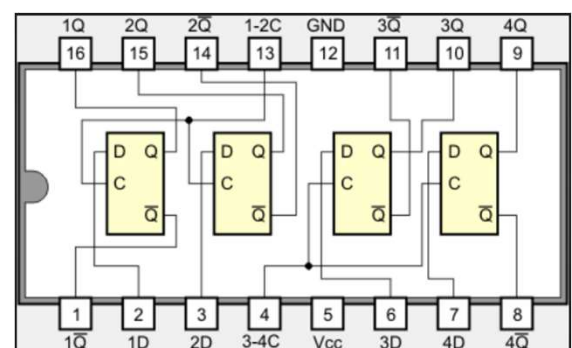
## 3. D Latch

**Type:** A latch is a **basic memory element** — it stores 1 bit of data.

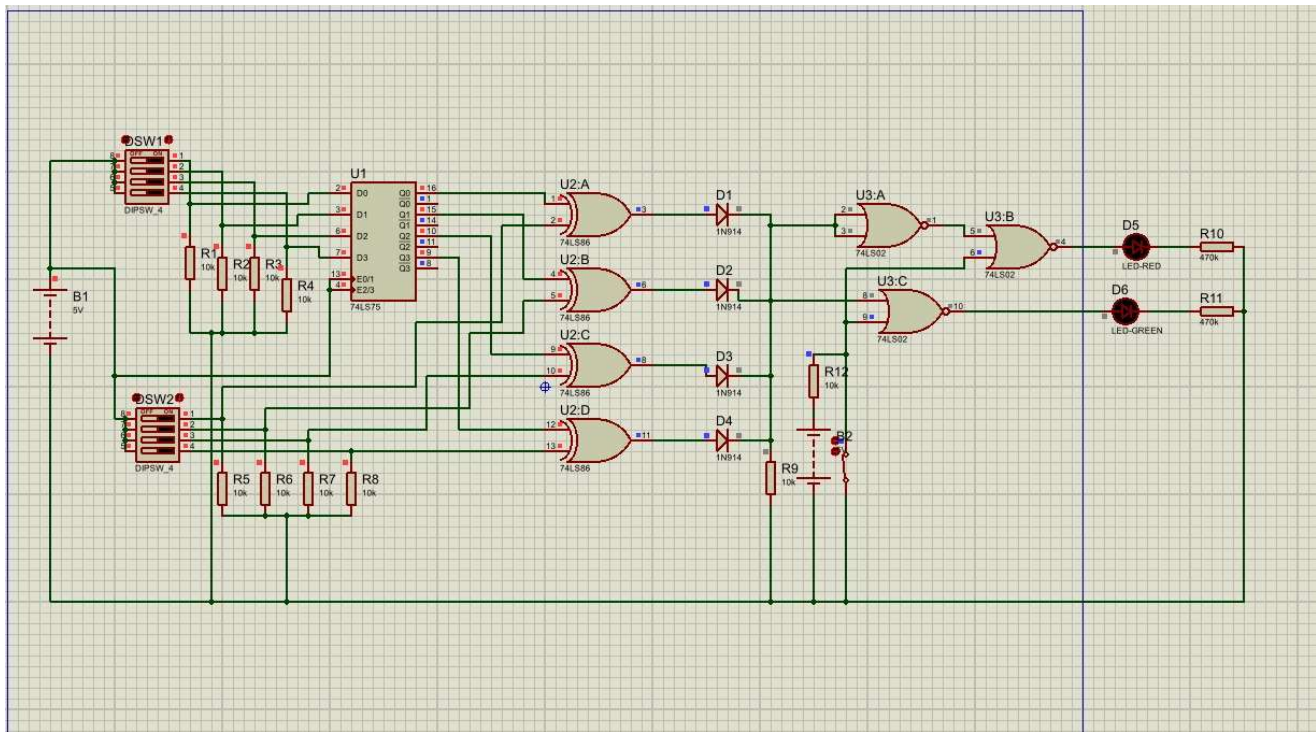
**Common Types:** SR Latch (Set-Reset), D Latch, JK Latch

**Example: SR Latch (using NOR gates)**

| D | G/EN | Q | Q' |
|---|------|---|----|
| 0 | 0    | Q | Q' |
| 1 | 0    | Q | Q' |
| 0 | 1    | 0 | 1  |
| 1 | 1    | 1 | 0  |



# Circuit Diagram



## Procedure

- Install the IC's , DIP Switches , resistor neatly on Breadboard .
- Wire the pin 7 & 14 of XOR and NOR IC's to Vcc and GND respectively.
- Connect pin 4 & 13 of Latch IC to each other and then to Vcc to control enable operation of Latch and pin 5 & 12 to Vcc and GND respectively.
- Wire one end of the DIP switchs with Vcc and other end to to Ground through 1K $\Omega$  resistor.
- Now connect the other end of one dip switch to input terminals pin 2,3,6,7 of Latch to store password in Latch.
- And connect second DIP switch other terminals to the XOR gate IC to pin 1,4,10,13 .
- Connect pin 16,15,10,9 of Latch to other terminal of XOR gate pin 2,5,9,12, respectively.

- viii. Now connect XOR gate output at one place through diode and connect it to pin 2,3,5 of NOR gate.
- ix. Now connect pin 6,8 of NOR to ground through switch. And output of first NOR gate from terminal 1 to pin 9 of it.
- x. Connect output of second and third NOR gate from pin 4 and 10 to the GREEN LED and RED LED respectively and other terminal of LED to ground through resistor.

## **Working Principle**

We are using one DIP switch to give stored/fixed password to Latch. When Latch Enable is HIGH then it gives output as input but when Enable is LOW then it gives output as per stored values irrespective of inputs.

Other DIP switch is used to enter the password which directly give to XOR as input and other input of XOR is from output of Latch. As we know XOR is used for comparator. This application of XOR we use here. If the enter password matches with stored password then XOR give LOW output and if not same then it gives HIGH output as per its Truth Table.

We connect all outputs of XOR gate at one pint through diode. The use of Diode here is to prevent reverse flow of current. As we have 4-bit pass word let 1<sup>st</sup> , 2<sup>nd</sup> and 3<sup>rd</sup> bit matched with stored data but 4<sup>th</sup> bit not so 4<sup>th</sup> XOR having output HIGH where as other have LOW so if we don't use diode then each expect 4<sup>th</sup> XOR others have same voltage at same point which not possible so we use diode so it get reverse biased in this condition and our circuit works correctly.

Now using NOR gate we glow our desired output as per correct or incorrect of password.

We use resistor at DIP switch to prevent floating values for latch and XOR gate input.

**BY:-**

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**124EI0019**